

# **Harmain Ali Butt 1781904958**

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Hiring Manager

Vercel

Re: Application for Lead Software Engineer

Hiring Manager,

I am writing to express my strong interest in the Lead Software Engineer position at Vercel. With over 6 years of experience as a seasoned Machine Learning and Artificial Intelligence (AI) engineer, I am confident in my ability to contribute significantly to your team and drive innovative solutions that align with Vercel's strategic objectives.

Throughout my career, I have consistently demonstrated a passion for developing and deploying cutting-edge AI models that deliver tangible business impact. As a Senior Machine Learning Engineer at AMD, I led the development of advanced AI models that optimized GPU performance, resulting in a 15% improvement in processing efficiency across key product lines. I also implemented robust quality control protocols and established automated monitoring systems to ensure the integrity of production data, reducing critical errors by 20%.

Prior to AMD, I served as a Machine Learning Engineer at NVIDIA, where I developed and deployed real-time image recognition and natural language processing models, improving data processing speeds by 25%. I also contributed to the development of custom ML/AI tools, integrating advanced statistics and algorithms for feature engineering and model evaluation.

Earlier in my career, I worked as an AI Developer at Google, where I developed core AI components for Google Cloud services, focusing on predictive analytics and anomaly detection. I also collaborated on web development projects, building user interfaces to expose ML model functionalities and insights to internal stakeholders.

With my strong technical expertise in Python, PyTorch, TensorFlow, C++, and a range of complementary technologies, I am well-equipped to tackle the challenges of the Lead Software Engineer role at Vercel. I am excited about the opportunity to leverage my skills and experience to drive innovation, enhance product performance, and contribute to the continued success of your organization.

Thank you for your consideration, and I look forward to the opportunity to discuss my qualifications further.

Sincerely,  
Harmain Ali Butt